Project 3: Final Report

Identification of Risk Factors and 10-year Stroke Risk in the Framingham Heart Study

Ryan Summers
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1. Introduction

The Framingham Heart Study (FHS) is a long-term prospective study investigating cardiovascular disease among community-dwelling individuals in Framingham, Massachusetts. This dataset includes up to 24 years of biennial follow-up for 4,434 participants, with repeated measures of clinical, laboratory, and questionnaire data, as well as adjudicated cardiovascular events.

The primary objective of this analysis is to identify baseline risk factors associated with incident stroke within 10 years and to estimate 10-year stroke probability across clinically relevant risk profiles. Analyses were stratified by sex to account for known differences in cardiovascular risk.

A secondary aim was to describe changes in key risk factors across three examination periods to assess whether a time-varying covariate approach may be warranted in future analyses.

The purpose of this report is to answer the research questions guiding this analysis: What baseline risk factors are significantly associated with stroke incidence within 10 years of follow-up, and how does the 10-year probability of stroke vary by risk profile and sex among participants in the FHS.

2. Methods

2.1 Data Cleaning

The original FHS dataset contained 4,434 participants with up to 24 years of follow-up structured in long format with repeated biennial visits per participant. Analyses were restricted to baseline (Exam 1) and the first 10 years of follow-up. Participants with prior stroke were excluded, yielding a final sample of 4,402 individuals.

The primary outcome is time to first stroke, measured in days, within 10 years. Participants without stroke were censored at 10 years or last follow-up. A binary indicator of stroke within 10 years (yes/no), was also created which will be used for computing 10-year stroke probabilities by risk profile.

Systolic blood pressure (**SBP**) and total cholesterol were rescaled so that the reported model coefficients correspond to a 10-mmHg and a 10-mg/dL increments, respectively, to improve interpretability.

2.2 Data Analysis

To address the primary research question, sex-stratified Cox proportional hazards models were used to identify risk factors associated with time to stroke. The proportional hazards assumption was assessed using Schoenfeld residuals.

Candidate baseline risk factors included age, SBP, and diabetes, selected a priori based on discussions with primary investigator (**PI**), with total cholesterol, BMI, smoking status, BP medication use, and prevalent cardiovascular disease (**CHD**) considered using backward stepwise selection based on corrected Akaike's Information Criterion (AICc), Schwarz Bayesian Criterion (SBC) criteria. Clinical relevance informed final model selection.

To address the secondary research question, 10-year stroke probabilities were estimated from Cox models for five prespecified risk profiles at ages 40, 50, and 60. The scenarios included a baseline profile with no major risk factors, high BP alone, diabetes alone, diabetes with high BP, and diabetes with high BP and a current smoker. Predicted risk was calculated as one minus the estimated 10-year stroke-free survival probability from the sex-specific Cox models.

As an additional descriptive aim, risk factors were also summarized across three examination periods using means (SD) and proportions to assess temporal changes.

All analyses used complete-case data under a missing at random assumption, where dropout depends on observed variables.

3. Results

Characteristics of the Framingham Heart Study Cohort

Characteristic	N	Overall N = 4,402[1]	Men N = 1,930[1]	Women N = 2,472[1]
Age	4,402	50 (9)	50 (9)	50 (9)
Systolic BP	4,402	133 (22)	132 (19)	134 (24)
Diabetes	4,402			
No		4,283 (97%)	1,871 (97%)	2,412 (98%)
Yes		119 (2.7%)	59 (3.1%)	60 (2.4%)
Total Cholesterol (mg/dL)	4,350	237 (45)	234 (42)	240 (46)
Missing		52 (1.2%)	7 (0.4%)	45 (1.8%)
Cardiovascular Disease	4,402			
No		4,215 (96%)	1,810 (94%)	2,405 (97%)
Yes		187 (4.2%)	120 (6.2%)	67 (2.7%)
BP Meds	4,342			
No		4,206 (97%)	1,868 (98%)	2,338 (96%)
Yes		136 (3.1%)	40 (2.1%)	96 (3.9%)
Missing		60 (1.4%)	22 (1.1%)	38 (1.5%)
Smoker	4,402			
No		2,233 (51%)	762 (39%)	1,471 (60%)
Yes		2,169 (49%)	1,168 (61%)	1,001 (40%)
BMI	4,385	25.8 (4.1)	26.2 (3.4)	25.6 (4.5)
Missing		17 (0.4%)	4 (0.2%)	13 (0.5%)
Event (Stroke)	4,402			
No-Censored		4,291 (97%)	1,881 (97%)	2,410 (97%)
Yes		111 (2.5%)	49 (2.5%)	62 (2.5%)

[1] Mean (SD); n (%)

Table 1. *Baseline characteristics of Framingham Heart Study participants, overall and stratified by sex during 10 years of follow-up.*

Baseline characteristics are shown in **Table 1.** Smoking and prevalent CHD were more common in men, while other risk factors were similar between sexes. Over 10 years, 111 stroke events occurred among 2,472 women and 1,930 men, corresponding to a cumulative incidence of approximately 2.5%.

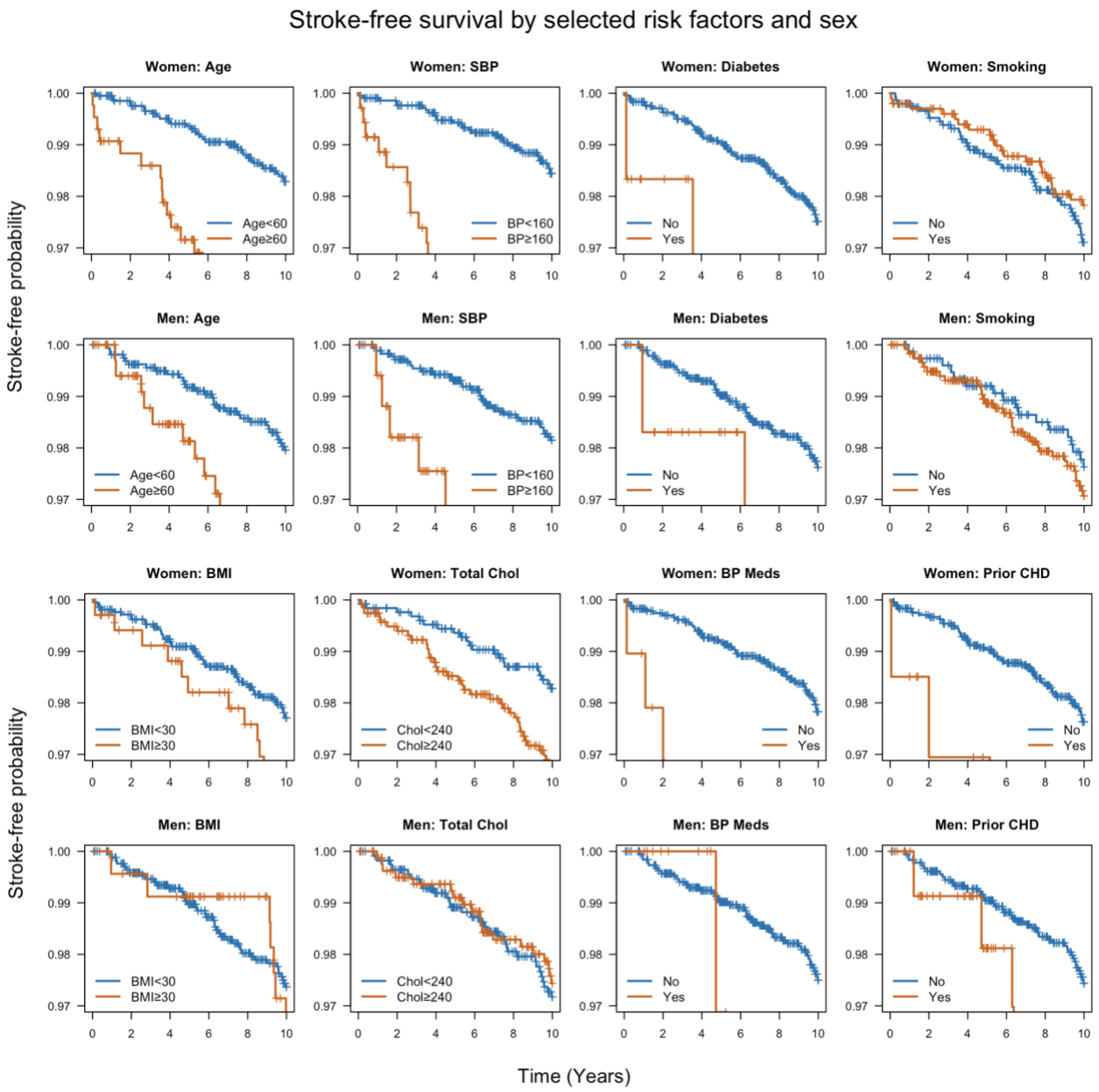

Figure 1. *Kaplan–Meier curves of stroke-free survival over 10 years, stratified by sex and baseline risk factors. Each panel displays survival probabilities for levels of the corresponding risk factor for women and men, with tick marks denoting censored observations.*

Kaplan–Meier curves, stratified by selected baseline risk factors, showed lower stroke-free survival among individuals with elevated SBP (≥160mmHg), diabetes, older age (≥60yrs), and prevalent CHD.

3.1 Risk Factors of Stroke

Stepwise procedures identified smoking as the only statistically significant predictor of stroke. However, prior literature, including the Framingham Stroke Risk Profile (Wolf et al., 1991), incorporated an interaction between SBP and BP medication use, as well as prevalent CHD, based on clinical relevance. These terms were therefore retained in the final sex-specific Cox models.

Hazard Ratios and Percent Change in Stroke Hazard by Sex

	Women (N=2378)		Men (N=1897)	
Risk factor	**HR (95% CI;P-val)**	**%Δ in Hazard**	**HR (95% CI;P-val)**	**%Δ in Hazard**
Age (per 1 year)	1.08 (1.03, 1.12; p=<0.01)	+7.5%	1.07 (1.03, 1.11; p=<0.01)	+6.6%
Systolic BP (per 10 mmHg)	1.28 (1.16, 1.41; p=<0.01)	+28.0%	1.41 (1.26, 1.59; p=<0.01)	+41.4%
BP meds (Yes vs No)	1.82 (0.03, 126.56; p=0.78)	+81.7%	+[1]	+[1]
Diabetes (Yes vs No)	1.81 (0.65, 5.05; p=0.25)	+81.4%	5.03 (2.18, 11.61; p=<0.01)	+403.4%
Current smoker (Yes vs No)	1.45 (0.83, 2.55; p=0.19)	+45.2%	1.97 (1.05, 3.67; p=0.034)	+96.5%
Cardiovascular Disease (Yes vs No)	1.65 (0.69, 3.94; p=0.26)	+65.3%	1.02 (0.36, 2.95; p=0.964)	+2.4%
SBP × BP meds interaction	0.99 (0.79, 1.25; p=0.93)	-1.0%	0.65 (0.44, 0.97; p=0.036)[1]	-35.0%[1]

[1] † Because the BP × BP medication interaction was significant in men, the effect of BP meds depends on BP and is not summarized as a single hazard ratio.

Table 2. *Sex-specific Cox proportional hazards models for 10-year stroke risk. Hazard ratios (95% CI; p-value) and percent change in hazard are shown for each risk factor. Continuous variables are scaled as indicated, and categorical variables are compared to their reference groups.*

In sex-stratified Cox models, age and SBP were the primary predictors of stroke in both sexes. SBP was associated with a 28% increase in hazard per 10 mmHg in women and 41% in men.

Among men, diabetes, smoking status, and the interaction between SBP and BP medication use were also important predictors. Diabetes was associated with a markedly increased hazard (HR ≈ 5), although the small number of events (N = 7) suggests some uncertainty in this estimate.

Current smoking was also associated with an approximately twofold increase in hazard (HR ≈ 2). The interaction indicates that men on BP medication have lower hazards at higher SBP levels compared to men not on BP medication.

Other predictors, including smoking, BP medication use, and prevalent CHD, were not statistically significant, likely reflecting limited power due to the relatively small number of stroke events within each stratum. For example, there were only seven events among women and five among men with prevalent CHD. Although CHD is a well-established risk factor for stroke, it was not statistically significant in this analysis.

3.2 10-year Probability of Stroke by Risk Profile

The estimated 10-year probabilities of stroke were low for individuals without any major risk factors, reflecting the relatively low incidence of stroke in this cohort over a 10-year period. Predicted risk increased with age and the presence of risk factors in both men and women. The highest risks were observed among individuals with multiple risk factors, particularly for men, where having diabetes, elevated BP, and being a current smoker resulted in the highest probabilities (**Table 3**).

Predicted 10-Year Probability of Stroke by Risk Profile and Sex

	Age 40		Age 50		Age 60	
Risk profile	**Women**	**Men**	**Women**	**Men**	**Women**	**Men**
Baseline	0.4%	0.3%	0.8%	0.7%	1.7%	1.2%
High BP	1.1%	1.4%	2.2%	2.6%	4.4%	4.8%
Diabetes	0.7%	1.7%	1.5%	3.3%	3.0%	6.1%
Diabetes + High BP	1.9%	6.8%	3.9%	12.4%	7.9%	22.1%
Smoker + Diabetes + High BP	2.8%	12.8%	5.6%	22.9%	11.2%	38.9%

Table 3. *Predicted 10-year stroke risk (%) by age, sex, and risk profile from sex-specific Cox models. Other risk factors were held constant at reference levels (absent of disease).*

3.3 Time-varying Covariates

Risk Factor Distributions Across Examination Periods by Sex								
	Men				Women			
Period	Age	SBP	Diabetes	Smoker	Age	SBP	Diabetes	Smoker
Exam 1	49.8 (8.7)	131.7 (19.4)	3.0%	60.4%	50.0 (8.6)	133.8 (24.5)	2.5%	40.4%
Exam 2	55.1 (8.5)	135.5 (19.9)	4.4%	52.0%	55.7 (8.6)	138.1 (24.3)	3.6%	37.8%
Exam 3	60.3 (8.2)	139.3 (21.2)	8.7%	38.9%	60.9 (8.4)	140.9 (24.1)	7.1%	31.0%

Table 4. *Distribution of risk factors across three examination periods in the Framingham Heart Study, stratified by sex. Continuous variables are presented as mean (SD), and categorical variables are presented as percentages. Estimates reflect cross-sectional summaries at each examination period.*

Risk factor distributions changed over time in both men and women. Systolic BP and diabetes prevalence increased over time, while smoking prevalence declined in both sexes.

4. Discussion

High SBP (hypertension) was identified as the primary risk factor for stroke in both sexes, consistent with findings from Wolf et al. Among men, smoking, diabetes and the interaction between SBP and BP medication use were also identified as important predictors of stroke.

Although several additional risk factors, were not statistically significan, their estimated effects were generally consistent with prior literature. The lack of statistical significance is likely attributable to the relatively low number of stroke events.

Predicted 10-year stroke risk increased with both age and accumulation of risk factors, with the highest probabilities observed among individuals with multiple concurrent risk factors.

Risk factor distributions were also observed to change across examination periods, with increases in SBP and diabetes prevalence and declines in smoking prevalence over time. These findings suggest that a time-varying covariate analysis may provide more accurate estimates in future work.

4.1 Limitations

Several limitations should be considered. First, the low number of stroke events limited statistical power, particularly for less prevalent risk factors and within sex-stratified analyses. Second, the use of baseline covariates may not capture changes in risk over time. Third, while model selection was guided by multiple information criteria, stepwise variable selection may introduce bias and overfitting, especially with correlated predictors. Finally, complete-case analysis may introduce bias if missingness is not at random.

4.2 Conclusions

In this cohort from the Framingham Heart Study, age and SBP were identified as the primary determinants of 10-year stroke risk in both men and women. While several additional risk factors were not statistically significant, their estimated effects were consistent with established clinical knowledge, suggesting that limited statistical power may have influenced inference.

Predicted stroke risk increased with the accumulation of risk factors, particularly among men. Observed changes in risk factor distributions over time further suggest that future analyses incorporating time-varying covariates may improve estimation of stroke risk.

References

ChatGPT was used to check R code errors for some of the packages used and to grammar check (OpenAI, 2025).

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Reproducibility Statement

The statistical code used in this analysis is available on my GitHub under the Code directory at: https://github.com/rysummers/BIOS6624-Class/tree/main/Project3

Raw data are not tracked in this repository and is available upon request.

Link to data on local drive: /Users/ryan_summers/GitHub/BIOS6624-Class/Project3/DataRaw/frmgham2.csv